

Qatar-2b

R-0961

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170404 · created 2026-07-29T16:06:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457847.89266 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.025
Transit depth (Rp/Rs)^2	1.88 +/- 0.69 [%]
Orbital Inclination (inc)	89.87 +/- 1.46
Airmass coefficient 1 (a1)	1.29 +/- 0.034
Airmass coefficient 2 (a2)	-0.166 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.6 %
Best Comparison Star	None
Optimal Aperture	8.81
Optimal Annulus	11.01
Transit Duration (day)	0.0747 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.025 vs 0.16327 (deviation 1.03 σ)
Duration fit vs published	0.0747 d vs 0.0767 d (deviation 0.97 σ)
Mid-time O-C	1.1 min
Depth SNR	5.4
Residual scatter	2.6 %

Light curve

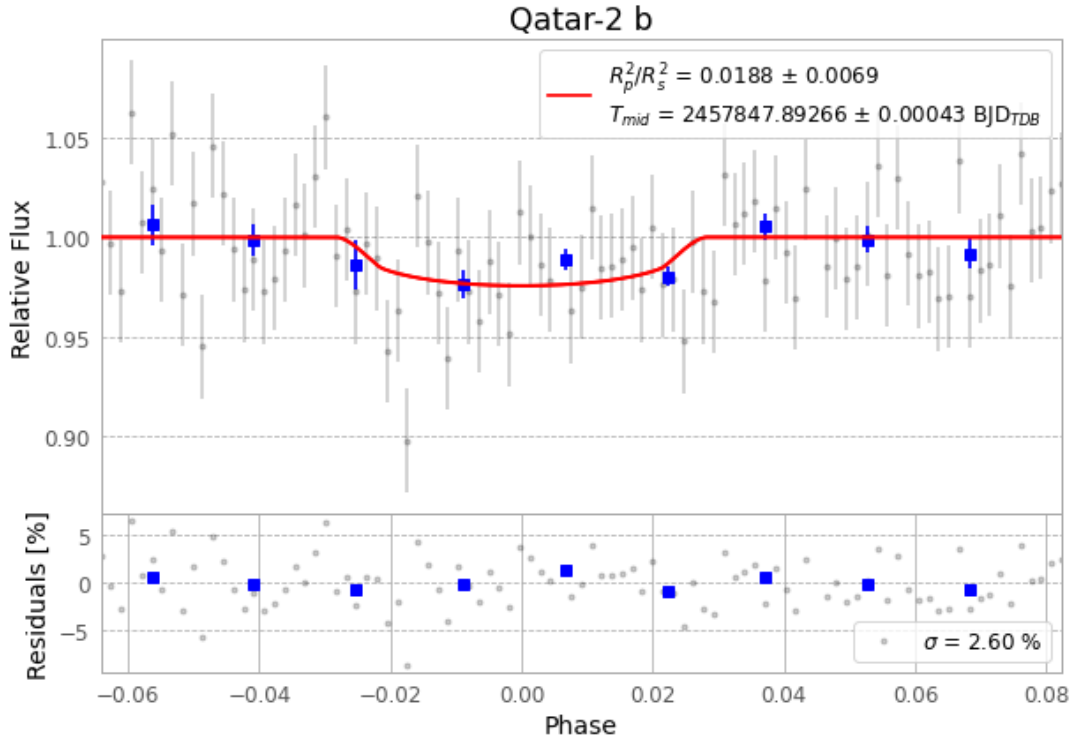


Figure 1 — FinalLightCurve_Qatar-2 b_2017-04-04.png (SHA-256 b42a9f9c020532ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 452], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a06a5df1ed113c8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

94 FITS frames (62 MB), Qatar-2170404071212.FITS → Qatar-2170404115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170404.log (23160acba6c...), FinalLightCurve_Qatar-2 b_2017-04-04.png (b42a9f9c0205...), FinalLightCurve_Qatar-2 b_2017-04-04.csv (cc9216406311...)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:06Z.